



Aluno:

QUESTION 1

(100 %)

Given an edge-weighted digraph, find a **monotonic shortest path** from s to every other vertex. A path is monotonic if the weight of every edge on the path is either strictly increasing or strictly decreasing.

QUESTION 2

(100 %)

Develop an implementation that can solve the longest-paths problem in edge-weighted DAGs.

QUESTION 3

(100 %)

Shortest path with the ability to skip one edge. Given an edge-weighted digraph with nonnegative weights, Design an $E \log V$ algorithm for finding the shortest path from s to t where you have the option to change the weight of any one edge to 0.

QUESTION 4

(100 %)

Widest Path Problem. Widest Path Problem is a problem of finding a path between two vertices of the graph maximizing the weight of the minimum-weight edge in the path.